Rajagopal Somaskandan

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Summary

IEEE-published Machine Learning Engineer and national hackathon winner (Industry Innovation '26) specializing in end-to-end AI systems, from fraud detection pipelines to RAG-based career platforms. Proficient in PyTorch, FastAPI, and LLM fine-tuning with production deployment experience via Docker and cloud-ready microservices. Seeking an entry-level ML/AI engineering role to build scalable, explainable AI systems.

Education

Sathyabama Institute of Science and Technology — 2025 – 2027 (In Progress)
M.Sc. Computer Science

Sathyabama Institute of Science and Technology — 2025
B.Sc. Computer Science (AI Specialization)

Technical Skills

Languages: Python, Java, SQL
Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Data Preprocessing, Cross Validation, Hyperparameter Tuning, Model Evaluation, Explainable AI (SHAP)
Deep Learning & NLP: PyTorch, TensorFlow, Neural Networks, CNNs, RNNs, LSTMs, Transformers, BERT, Sentence-Transformers
Generative AI & Retrieval: RAG, LangChain, Embeddings, Vector Databases, FAISS, ChromaDB, Semantic Search, Agentic AI, LLM Fine-Tuning (LLaMA)
Backend & APIs: FastAPI, REST APIs, Microservices
Data Engineering:EDA, Data Cleaning, Selenium, Pandas, NumPy
Computer Vision & 3D AI: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting, FFmpeg
Databases: PostgreSQL, MongoDB
Cloud, DevOps & MLOps: AWS Fundamentals, Docker, Kubernetes, Git, GitHub, GitHub Actions, CI/CD, MLflow, Model Deployment
Frontend: React, Tailwind CSS, Streamlit

Work Experience

Production AI Developer (Intern) — Mar 2026 – Jun 2026
InnoXR Labs Pvt. Ltd. — *Chennai, India*

- Developed an applied **machine learning pipeline** for single-video **3D reconstruction** (MonoSplat) using **COLMAP** for camera pose estimation and **Gaussian Splatting** in **PyTorch**.
- Worked on **AI** system design, building a hybrid **CPU-GPU data preprocessing pipeline** with **FFmpeg** for cloud-based execution.
- Deployed a scalable **FastAPI** backend with async job handling and a **Three.js** viewer for end-to-end 3D visualization, within an immersive technology startup environment.

Data Analyst Intern — Jul 2024 – Oct 2024
India Advocacy — *Chennai, India*

- Built Java and Selenium **ETL pipelines** to scrape and process **1M+ records**; performed **data preprocessing and cleaning** to transform raw data into **15–20k structured datasets** through multi-stage validation.

Security Trooper (National Service) — Jan 2020 – Jan 2022
Singapore Armed Forces — *Singapore*

- Executed security and vehicular operations in regulated environments, ensuring strict protocol compliance and operational readiness.
- Supported COVID-19 response initiatives through structured data collection and reporting under time-sensitive conditions.

Sales Associate (Part-Time) — Feb 2022 – Aug 2022
Bata — *Singapore*

- Handled high-volume customer interactions in a fast-paced retail environment, ensuring efficient service delivery.
- Managed billing, inventory coordination, and issue resolution to support smooth store operations.

Projects

FinGuard Pro: Real-Time Fraud Detection & Compliance System IEEE ICAAIC 2025

- Built a UPI fraud detection system trained on **100,000 synthetic transactions** using **25+ engineered features** including geo-deviation scores and amount-to-average ratios.
- Conducted **A/B experimentation** across **XGBoost**, **LightGBM**, and **CatBoost** with **hyperparameter tuning** via **Optuna TPE**; **XGBoost** selected (**94% recall**, **ROC-AUC 0.9951**, **F1 0.57**) over LightGBM (F1 0.56) and CatBoost (F1 0.48).
- Integrated **SHAP TreeExplainer** for global/local interpretability and a **fuzzy AML screening engine** (RapidFuzz, 5,000-entity watchlist) with rule-based detection of structuring, round-tripping, and velocity anomalies.
- Deployed a dual-dashboard (**FastAPI + Streamlit**) with real-time alerts, SHAP visualizations, and automated PDF/ZIP compliance report generation; **inference optimization** achieving sub-**3-second** latency per transaction.

CareerGenie: AI Career Platform Hackathon Winner, Industry Innovation '26

- Built an end-to-end **RAG pipeline** using **MiniLM embeddings** and **ChromaDB** for semantic job matching, replacing keyword-based retrieval with skill-aware vector search.
- Designed a **multi-agent orchestration system** (ReAct-style) with 5 specialized agents (Resume, Job, Interview, Roadmap, Advisor) and an **Agent Debate System** (3 proposers, critic, synthesizer) to reduce hallucinations.
- Engineered a **Learning-to-Rank engine** using **BPR** trained on click, application, and offer signals; integrated **SerpAPI + Google Trends** with weighted scoring (semantic 35%, skills 45%, title 20%).
- Performed **LLM fine-tuning** on **LLaMA** via multi-provider waterfall (Groq → Anthropic) improving response quality and reducing latency under degraded provider conditions.
- Delivered full-stack system: **React + Tailwind** frontend deployed on **Netlify**, **FastAPI** backend via **ngrok tunneling**, supporting ATS scoring, resume rewriting, mock interviews, and learning roadmaps.

MonoSplat: Single-Video 3D Reconstruction Pipeline Internship Project, 2026

- Engineered a 3D reconstruction pipeline using **COLMAP** and **Gaussian Splatting** in **PyTorch**.
- Designed a hybrid **CPU-GPU data pipeline**, optimizing **data preprocessing** with **FFmpeg** for cloud execution.
- Built an async **FastAPI** backend with a **Three.js** viewer for interactive 3D visualization.
- Focused on performance and usability, enabling smooth rendering and faster end-to-end processing.

S.A.F.E: Privacy-Preserving Crowd Anomaly Detection Design Thinking Project, 2026

- Developed a privacy-first crowd anomaly detection system using **Farneback optical flow** and a **2×3 zone grid** to extract density, velocity, and direction variance metrics without tracking individuals.
- Conducted **A/B experimentation across 5 unsupervised models** (MAD, Z-Score, One-Class SVM, Isolation Forest, LSTM Autoencoder) trained on **15,928 zone records** (ETH/UCY); validated on **13,805 records** (Zara1/Zara2).
- **MAD** selected as primary model after cross-domain evaluation on the Mall dataset (**ROC-AUC 0.8391**, **F1 0.4130**); reduced false positives by **40–60%** via adaptive camera shake detection (threshold 0.25, 3-frame check).
- Built a **FastAPI** backend and **Streamlit** dashboard with explainable real-time alerts and zone-level reasoning (e.g., "Density >50% in Zone 3") for safety operators.

SourceUp: Explainable Supplier Recommendation System Final Year Capstone, 2026

- Implemented semantic supplier search using **Sentence-BERT** and **FAISS** for natural language queries.
- Designed a ranking model using **Scikit-learn** with cost, delivery time, and reliability constraints.
- Added an explainability layer to provide feature-level insights for supplier selection and risk assessment.

Publications & Conferences

FinGuard Pro: Explainable AI for Financial Fraud Detection | *ICAAIC 2025* ISBN: 979-8-3315-6587-9

Achievements & Hackathons

Industry Innovation Hackathon | *Sathyabama Institute of Science and Technology – National Level* Mar 2026

- Won **Best Industry Innovation Award** at Industry Innovation '26.

Origin 24-Hour Hackathon | *SIMATS Engineering, Chennai* Apr 2026

- Placed **4th** at Origin 24-Hour Hackathon.

Certifications

AI & ML | *Space Zee Technologies* Jul 2024
Python Core | *Space Zee Technologies* Feb 2024